

Role 2 — Protocol & Addressing Analyst

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Owens: protocol analysis table (layer, protocol — DNS/HTTP/TLS/TCP/IP/Ethernet, PDU, addressing field used, purpose) and the explanation of IP addresses (logical, routable), MAC addresses (physical, local-hop only), and port numbers (identify the application/service — 53 for DNS, 443 for HTTPS).

SLIDE 7 — Protocol Analysis Table

Layer	Protocol	PDU	Addressing Info	Purpose
Application	DNS	Data (message)	Domain name → IP address	Resolves the website's domain name to its IP address before anything else can happen
Application	HTTP over TLS (HTTPS)	Data (message)	None (relies on ports below)	Requests the web page and carries the encrypted response back from the server
Transport	TCP	Segment	Source port (ephemeral) / Destination port (443, or 53 for DNS-over-TCP)	Reliable, ordered, connection-based delivery between the two applications; opens the connection via the 3-way handshake
Internet	IP	Packet	Source IP / Destination IP	Logical addressing and routing — lets routers move the packet across different networks toward the destination
Network Access	Ethernet / Wi-Fi	Frame	Source MAC / Destination MAC	Physical addressing and delivery across one local link; re-applied at every hop

This table is our map of every protocol involved in opening a secure website, laid out by which TCP/IP layer it belongs to. At the Application layer we have DNS, which resolves the domain name, and HTTP wrapped in TLS, which is the actual encrypted request for the page. At the Transport layer, TCP is what carries that request reliably, using source and destination ports. At the Internet layer, IP handles logical addressing — the IP addresses that let routers find the destination network. And at the Network Access layer, Ethernet handles the physical addressing — MAC addresses — for each local hop. Each layer only needs to understand its own protocol and its own addressing scheme; that's what lets the whole stack work together without any one layer needing to know how the others do their job.

SLIDE 8 — IP Addresses and MAC Addresses

On-slide text:

- IP address: logical, software-assigned, routable across networks — identifies where a device sits on the internet
- Static vs dynamic IP: manually fixed vs auto-assigned each session via DHCP
- MAC address: physical, burned into the network card, only meaningful on the local (one-hop) segment
- Format: 48-bit, 12 hex characters (e.g. 00:1A:2B:3C:4D:5E) — first half identifies the manufacturer (OUI), second half the specific device

It helps to think of IP and MAC addresses as answering two different questions. The IP address answers 'which network, and where on the internet' — it's a logical address, it can be static, fixed once by an administrator, or dynamic, handed out automatically by DHCP each time a device connects, and it's what makes a device routable, meaning routers across the internet can use it to find the right path. The MAC address answers a completely different question: 'which physical device, on this one

local link.' It's burned into the network card at manufacture, it's a 48-bit number written as 12 hexadecimal characters, and the first half identifies the manufacturer while the second half is unique to that device. And critically — unlike the IP address, which stays the same for the whole journey, the MAC address only means something for a single physical hop; it gets replaced every time the packet is re-framed by the next router, which is exactly what Emmanuel's diagram showed.

SLIDE 9 — Port Numbers in Our Scenario

On-slide text:

- Port 53 — DNS: resolves domain names to IP addresses; runs over UDP (or TCP for larger responses)
- Port 443 — HTTPS: encrypted web traffic over TCP, secured with TLS
- Ports identify the specific application/service on a device — not the device itself, that's the IP address's job

Port numbers solve one more problem: once a packet has arrived at the correct device using its IP address, how does the operating system know which application on that device it's for? That's what the port number is for. In our scenario we see two of the most important ports on the internet: port 53, used by DNS to resolve the domain name we typed into an IP address, and port 443, used by HTTPS for the actual encrypted page request and response, secured using TLS. Port 443 specifically tells the receiving system, and any firewall in between, that this is expected to be an encrypted session, without needing to state that anywhere in the URL — that's why we just type `https://` and it works

. References

- SecureW2 — "What Is Port 443?" — <https://securew2.com/blog/what-is-port-443>
- EfficientIP — "What Is a DNS Port?" — <https://efficientip.com/glossary/what-is-dns-port/>
- SSL Wiki (Medium) — "What Is HTTPS Port 443?" — <https://medium.com/@sslwiki/what-is-https-port-443-59a0298260b0>
- Cloudflare Learning Center — "What Is the Internet Protocol?" — <https://www.cloudflare.com/learning/network-layer/internet-protocol/>
- NetworkWalks — "IP Address and Its Types" — <https://networkwalks.com/ip-address-its-types/>